HDOC Day-38 — String Operations
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Algorithm

1. Start.
2. Declare a character array to store the string.
3. Read the string using `fgets()`.
4. Repeatedly display the menu:
   ◦ Count spaces, digits, and special characters.
   ◦ Count frequency of a given character.
   ◦ Toggle case of each character.
   ◦ Exit.
5. Read the user's choice.
6. Use a `switch` statement:
   ◦ **Choice 1:** Call a function that examines every character and counts spaces, digits, and special characters.
   ◦ **Choice 2:** Read the character to search for and call a function to count its occurrences.
   ◦ **Choice 3:** Call a function that converts lowercase letters to uppercase and uppercase letters to lowercase.
   ◦ **Choice 4:** Exit the program.
   ◦ Otherwise, display `"Invalid choice!"`.
7. Continue until choice 4 is selected.
8. Stop.

Test-Case Table

Test Case	Operation	Input	Expected Output
	1 Count spaces/ digits/ special	Hello 123!	Spaces = 1, Digits = 3, Special = 1
	2 Count frequency	String: banana, Character: a	Frequency = 3
	3 Toggle case	Hello World	hELLO wORLD
	4 Count spaces/ digits/ special	C Programming @ 2026	Spaces = 3, Digits = 4, Special = 1
	5 Count frequency	String: programming, Character: m	Frequency = 2
	6 Toggle case	AbC123@xYz	aBc123@XyZ
	7 Invalid menu choice	Choice = 6	Invalid choice!
	8 Exit	Choice = 4	Program terminates

C Program

```
#include <stdio.h>
void countCharacters(char str[]);
void characterFrequency(char str[]);
void toggleCase(char str[]);
int main()
{
char str[200];
int choice;
printf("Enter a string: ");
fgets(str, sizeof(str), stdin);
do
{
printf("\n----- MENU -----\n");
printf("1. Count spaces, digits and special
characters\n");
printf("2. Count frequency of a given character\n");
printf("3. Toggle the case of each character\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1:
countCharacters(str);
break;
case 2:
characterFrequency(str);
break;
case 3:
toggleCase(str);
break;
case 4:
break;
printf("Exiting program...\n");
```

```
default:
printf("Invalid choice! Please enter 1 to
4.\n");
}
} while (choice != 4);
return 0;
}
/* Function to count spaces, digits and special characters */
void countCharacters(char str[])
{
int i = 0;
int spaces = 0, digits = 0, special = 0;
while (str[i] != '\0')
{
if (str[i] == ' ')
{
spaces++;
}
else if (str[i] >= '0' && str[i] <= '9')
{
digits++;
}
else if (!((str[i] >= 'A' && str[i] <= 'Z') ||
(str[i] >= 'a' && str[i] <= 'z')) &&
str[i] != '\n')
special++;
{
}
i++;
}
printf("\nNumber of spaces = %d\n", spaces);
printf("Number of digits = %d\n", digits);
printf("Number of special characters = %d\n", special);
}
/* Function to find frequency of a character */
void characterFrequency(char str[])
{
char ch;
```

```
int i = 0, count = 0;
printf("Enter the character to search: ");
scanf(" %c", &ch);
while (str[i] != '\0')
{
if (str[i] == ch)
{
count++;
}
i++;
}
printf("Frequency of '%c' = %d\n", ch, count);
}
/* Function to toggle case */
void toggleCase(char str[])
{
int i = 0;
printf("String after toggling case: ");
while (str[i] != '\0')
{
if (str[i] >= 'a' && str[i] <= 'z')
{
printf("%c", str[i] - 32);
}
else if (str[i] >= 'A' && str[i] <= 'Z')
{
printf("%c", str[i] + 32);
}
else
{
printf("%c", str[i]);
}
i++;
}
}
```